Md. Tanvir Hasan Anik

CLOUD & PLATFORM ENGINEER

Senior Software Engineer — Cloud Infrastructure · Backend Systems · Linux Operations

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github.com/ProAnik linkedin.com/in/md-tanvir-hasan-anik

// EXECUTIVE PROFILE

Software engineer with 5+ years of experience engineering high-availability cloud infrastructure and mission-critical backend systems. Specialized in codifying multi-environment AWS platforms via Terraform/Terragrunt, scaling MongoDB replica sets on EC2, implementing deep observability in Datadog/CloudWatch, and deploying low-latency APIs in Python (FastAPI), C# (.NET), and PHP (Laravel). Experienced in full-lifecycle production ownership: network architecture, security boundaries, containerization, and on-call reliability.

// TECHNICAL EXPERTISE

› CLOUD & IAC

AWS (EC2, VPC, IAM, Route 53, RDS, S3, Secrets Manager, WorkSpaces, CloudWatch), Terraform, Terragrunt, Docker, Kubernetes, JFrog Artifactory, GitHub Actions CI/CD, Azure

› LINUX & SECURITY

Linux administration (Ubuntu, Debian, RHEL), user & PAM management, StrongSwan IPsec VPN, firewalls (iptables / UFW), Nginx, Apache, Supervisor, SSL/TLS, VPC networking

› BACKEND & APIS

Python (FastAPI), C# (ASP.NET Core), PHP (Laravel), C++, RESTful APIs, gRPC, WebSockets, Node.js / Express, React, Next.js, Vue.js

› DATA & MONITORING

MongoDB (multi-node replica sets), PostgreSQL, MySQL, MSSQL, Redis (caching & queues), Datadog (APM & tracing), AWS CloudWatch, Databricks

// PROFESSIONAL EXPERIENCE

Full Stack Developer — Admissions Platform & Cloud Delivery · BHE Uni

2025 - Present

University admissions CRM, lead lifecycle, and campus enrollment infrastructure

- **Automated CI/CD & Release Engineering**: Orchestrated automated GitHub Actions pipelines, Docker containerized releases, Nginx reverse proxying, automated SSL/TLS rotation, and zero-downtime rollback deployments on Linux production servers.
- **High-Concurrency Backend & Queue Infrastructure**: Scaled high-throughput ingestion using Laravel Horizon workers, scheduled daemons, Redis distributed caching, and indexed SQL queries, maintaining sub-50ms p95 latency during peak admissions.
- **Resilient Webhook Listeners & CRM Workflows**: Engineered idempotent Meta Graph API listeners for real-time lead capture, automated applicant scoring heuristics, and modular CRM features (FastAPI, Laravel, Next.js, React) with audit trails and RBAC.
- **Secure Applicant Document Pipeline**: Architected secure document upload and verification workflows for academic transcripts and identity records, utilizing encrypted AWS S3 storage with presigned temporary URLs to enforce strict privacy.
- **Platform Delivery & Engineering Mentorship**: Delivered student portals (bheuni.uk, ambassador.bheuni.uk, bheuni.io, bangladesh247.com); coordinate Agile sprints and conduct code reviews.

Program Manager / Senior Software Engineer · D2Soft Intelligence

2022 - 2025

Cloud infrastructure, distributed backend systems, enterprise and aerospace platforms

- **Infrastructure as Code & Registry Automation**: Managed multi-tier AWS environments entirely as code using Terraform and Terragrunt across VPCs, EC2, IAM, and networking; integrated JFrog Artifactory for container images and binary artifact lifecycles.
- **VPC Networking, Security & Developer Cloud**: Architected AWS VPCs with public/private subnets, NAT Gateways, Route 53, and cross-VPC peering; secured credentials in AWS Secrets Manager and provisioned isolated AWS WorkSpaces developer environments.
- **Scaled MongoDB Replica Sets on EC2**: Architected and scaled self-hosted MongoDB multi-node replica sets on EC2 across multiple Availability Zones with automated primary failover, EBS snapshot schedules, and private-subnet isolation.
- **Full-Stack Observability & Linux Fleet Administration**: Configured Datadog APM and CloudWatch alarms reducing MTTR; hardened Linux fleets with PAM, iptables/UFW, Nginx/Apache reverse proxies, and StrongSwan IPsec site-to-site VPN tunnels.
- **Aerospace Compute, APIs & US Marketplace**: Delivered performance-critical C++ compute modules for government aerospace contracts; built distributed backend microservices (FastAPI, ASP.NET Core) powering bakome.com US food

// EXPERIENCE — BANGLA PUZZLE LIMITED

Web Developer — Government & Public Sector Systems · Bangla Puzzle Limited

2021 – 2022

National government e-commerce, healthcare telemetry, and enterprise web platforms

- **Linux Hosting & Server Operations**: Administered Linux production hosting environments, configured Nginx and Apache reverse proxies, scheduled cron jobs, and managed SSL certificates with zero-downtime release deployments.
- **sadai.gov.bd National E-Commerce API**: Built the core e-commerce REST API backend for sadai.gov.bd (Department of Agricultural Marketing, Ministry of Agriculture), connecting farmers, wholesalers, and consumers directly across web and two Android apps (Sadai and Sadai Uddokta).
- **DGFP Field Monitors Telemetry Backend**: Developed the backend for the Directorate General of Family Planning's field monitoring platform (Ministry of Health & Family Welfare), used by field officers across all eight divisions of Bangladesh for health inspection reporting.
- **Fintech & Secure Payment Integrations**: Integrated bKash and SSLCommerz payment gateways featuring cryptographic webhook verification, ledger reconciliation, and automated receipts for Laravel, Vue.js, and React web apps.

// SELECTED FLAGSHIP ARCHITECTURE PROJECTS

sadai.gov.bd — National Agri E-Commerce GOV PLATFORM

Architected the multi-app REST API handling farmer-to-consumer transactions, product catalogs, and bKash/SSLCommerz payments for the Ministry of Agriculture.

Laravel · MySQL · Redis · Nginx · Linux

DGFP Field Monitoring Platform NATIONAL HEALTH

Engineered the data synchronization backend collecting and aggregating maternal and field health monitoring telemetry across all 8 divisions of Bangladesh.

Laravel · PostgreSQL · REST API · Cron

bakome.com Multi-App Delivery Platform US MARKETPLACE

Engineered backend APIs and order routing powering customer, courier bearer, and vendor merchant apps for an international food delivery marketplace in MD, DC, and VA.

FastAPI · ASP.NET Core · Docker · AWS

AWS Multi-VPC & Scaled MongoDB Clusters CLOUD & IAC

Automated AWS multi-VPC provisioning with Terraform and Terragrunt; scaled multi-node MongoDB replica sets on EC2 with Datadog APM and CloudWatch alarms.

Terraform · Terragrunt · MongoDB · Datadog

// EDUCATION & RESEARCH

M.Sc. in Computer Science · 2024 – Present

Jahangirnagar University · Research: AI applications in medical science

B.Sc. in Computer Science & Engineering · 2017 – 2020

Daffodil International University · Core: Distributed Systems, Networks, OS

// TRAINING & METHODOLOGIES

Network & Linux System Administration

Genuity Training · IPsec VPNs, firewalls, routing, user permissions

Engineering Methodologies & Languages

Agile / Scrum, Jira, Postman, Git · Bengali (Native), English (Professional)